Venkata Lakshmi Parimala Pasupuleti

AI Solution Architect · Agentic Systems · LLM Inference API Integration · Multi-Agent Orchestration · Open Source · Python · FastAPI · RAG · 9yr Production

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PROFESSIONAL SUMMARY

AI Solution Architect with 9+ years of production engineering and 2+ years shipping customer-embedded agentic AI systems - the exact profile FriendliAI's Solution Architect role is built around. At Magizhs, I was the sole technical owner embedded with an external customer: translated ambiguous requirements into a production LangGraph agentic system from discovery through inference API integration, workflow orchestration, and deployment in weeks. AgentProbe is the open-source proof point: a multi-provider LLM inference routing platform (Anthropic, OpenAI, Ollama) with a 15-endpoint FastAPI backend, 5-metric evaluation engine, pass@k multi-trial harness, and CI regression gate - the kind of agentic developer infrastructure FriendliAI builds for its customers. LeaseI demonstrates the multi-agent production system FriendliAI APIs must power: 8-agent LangGraph StateGraph (workflow DAG) with RAG pipelines, memory management across sessions, tool invocation, and HITL guardrails achieving 93.6% precision across 1,240 test cases. I build with Python, FastAPI, and LangChain daily; I integrate and extend open-source generative AI frameworks; and I deliver technical workshops and production guidance to non-technical stakeholders. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic Systems & LLM Integration: Python (9yr), FastAPI, LangChain, LangGraph, MCP, Multi-Agent Orchestration, Workflow DAGs, Inference API Integration, LLM Inference, Open-Source AI Frameworks, Agent Frameworks, Generative AI APIs, Anthropic API, OpenAI API, Ollama, Multi-Provider LLM Routing, Prompt Engineering, API Composition, Workflow Orchestration, Context Engineering, Memory Management, Tool Calling, Agent Debugging, Output Validation, HITL Guardrails

RAG & Knowledge Systems: RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, Embeddings, Knowledge Retrieval, Document Intelligence, Evaluation Frameworks, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, CI Regression Gate, Observability, Tracing, Feedback Loops, SDK Development, REST APIs, API Design, Open Source

Infrastructure & Data: AWS (S3, EC2, RDS, Glue, Lambda), Serverless Backends, API Gateways, Docker, Microservices, RabbitMQ, PostgreSQL, MySQL, Oracle, Redis, React.js, TypeScript, GitHub Actions CI/CD, Python Automation, Linux, Git, Agile/Scrum, Technical Workshops, Developer Experience, Cross-functional Delivery

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Customer-Embedded Agentic AI Delivery

- Embedded with external customer as sole technical owner - led technical discovery, designed multi-agent LangGraph workflow orchestration, integrated LLM inference APIs (Anthropic, OpenAI), implemented RAG retrieval layer over operational knowledge corpus, built HITL validation guardrails for regulated decision paths, and delivered production deployment with observability and feedback loop infrastructure in weeks. Direct customer-embedded delivery: requirements → architecture → inference integration → production.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Delivered technical workshops to 600+ students in Python, SQL, and data engineering; built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards for faculty leadership (40% comprehension improvement); cross-functional technical delivery to non-technical stakeholders across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Backend & API development at scale** - Python, FastAPI/Flask REST APIs, PostgreSQL/MySQL/Oracle, RabbitMQ async event processing with retry logic and idempotency guarantees, AWS S3/EC2/RDS/Glue, Docker microservices redesign (50% scalability improvement); owned production systems for 7 years at 99.5% uptime across 100K+ housing units; 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD.
- **Developer experience & platform engineering** - built internal debugging tools reducing triage time 40%; Python/Bash automation eliminating 80% of manual workload; 300+ peer code reviews; mentored 5+ engineers on system design and production practices; wrote design docs and architecture decisions; delivered technical guidance across product, compliance, and operations stakeholders in 120+ Agile sprints.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Open-Source Agentic AI Evaluation Platform

Open Source · Multi-Provider LLM Routing · FastAPI · RAG · CI Gate

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Open-source AI evaluation and inference routing platform - the kind of agentic developer infrastructure FriendliAI delivers to its customers. **Multi-provider LLM inference routing:** Anthropic, OpenAI, Ollama - pluggable provider architecture with cost and latency tradeoffs; custom inference endpoints for evaluation workflows. **Evaluation engine:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness surfacing variance in non-deterministic outputs. **Platform:** 15-endpoint FastAPI backend; MCP integration; CI-integrated regression gate (GitHub Actions); offline batch evaluation pipelines. Open-source, enterprise-deployable. Zero regressions across 1,240+ test cases.

Leasel: Production Multi-Agent System - Workflow DAG + RAG + Memory

Multi-Agent · LangGraph · RAG · Memory Management · HITL · Production

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system demonstrating the agentic architecture FriendliAI APIs power at scale. **Workflow DAG:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing based on confidence scores - a multi-step agentic workflow DAG for automated investigation. **RAG & memory:** SQLite knowledge repository persisted across all agents and sessions - context and memory management across long-running agent sessions. **Tool integration:** agents invoke retrieval, structured data, and configuration tools with grounded, citation-backed outputs. **Guardrails:** deterministic HITL escalation blocks unsafe decisions. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Agentic System - LLM Inference API Integration

Full Stack · API Integration · HIPAA · Serverless · AWS

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · BERT · spaCy · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack agentic system with LLM inference API integration - natural-language queries translated into structured FHIR API calls via LLM tool invocation; HIPAA-compliant RBAC with full data isolation and audit trail; serverless backend on AWS Lambda with API gateway integration; ~60% reduction in manual retrieval time. Demonstrates production LLM inference integration into enterprise APIs with regulated data requirements.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017